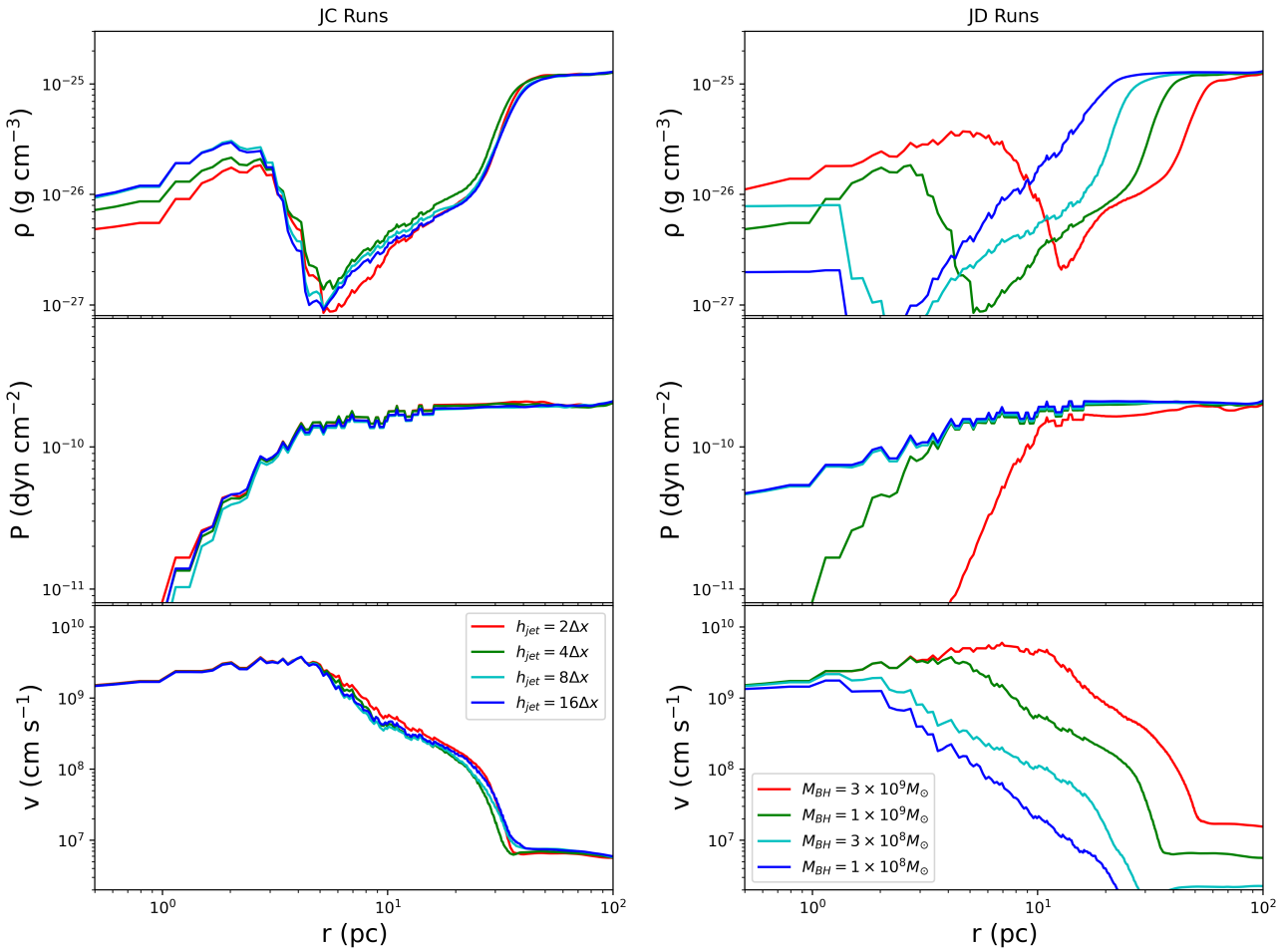

Figure 17. Figure 12 continued. Left: With fixed resolution and jet radius, varying jet injection height (JC runs). Right: Varying the mass of the central BH and fixing other parameters (JD runs).

and is observed in our simulations. The KHI contributes to the formation of the small-scale turbulence effects at the jet-ICM interface. Through reconfinement, the jets also develop centrifugal instability (CFI) that generates turbulent flow (Gourgouliatos & Komissarov 2018).

Many kinds of instabilities are seen in our jet runs, especially with short, wide jets, which is similar to what Monceau-Baroux et al. (2012) has observed: a jet with a wider opening angle decelerates faster and produces a wider radial expansion zone dominated by instabilities. The Kelvin-Helmholtz instability is the most prominent since it is expected to occur in pure hydrodynamic simulations where there is a velocity shear.

Some other signatures of our simulated jets, including shock structure and energy transfer behavior, are similar to predictions from analytical models and previous simulations of the interaction of jets with the cluster-environment gas (e.g., Begelman & Cioffi 1989). Our jets show a nose-cone structure as well as several other structures that represent various instabilities seen in MHD jet simulations (e.g., Mignone et al. 2010). Compared with works such as Perucho et al. (2019), the ratio of the width of the jet shocked region to the distance from the BH ranges from about 200 (closer to the BH) to 20 (edge of the jet) in our simulations, which is similar to what they observed. The higher the jet power, the lower this ratio becomes, which is also consistent with the work mentioned above.

4.2. *Comparison and Implications for Larger Scale Cluster Simulations*

Based on our runs with different accretion and jet feedback configurations, we found certain differences with many galaxy cluster-scale simulations.

As indicated previously, most such simulations use the Bondi formula with a boost parameter to characterize accretion rates. Some models that aim to incorporate "cold-mode accretion" usually involve dividing the total amount of cold gas within the accretion zone by a certain timescale, or measuring the mass change within a few zones around the SMBH (such as Gaspari et al.

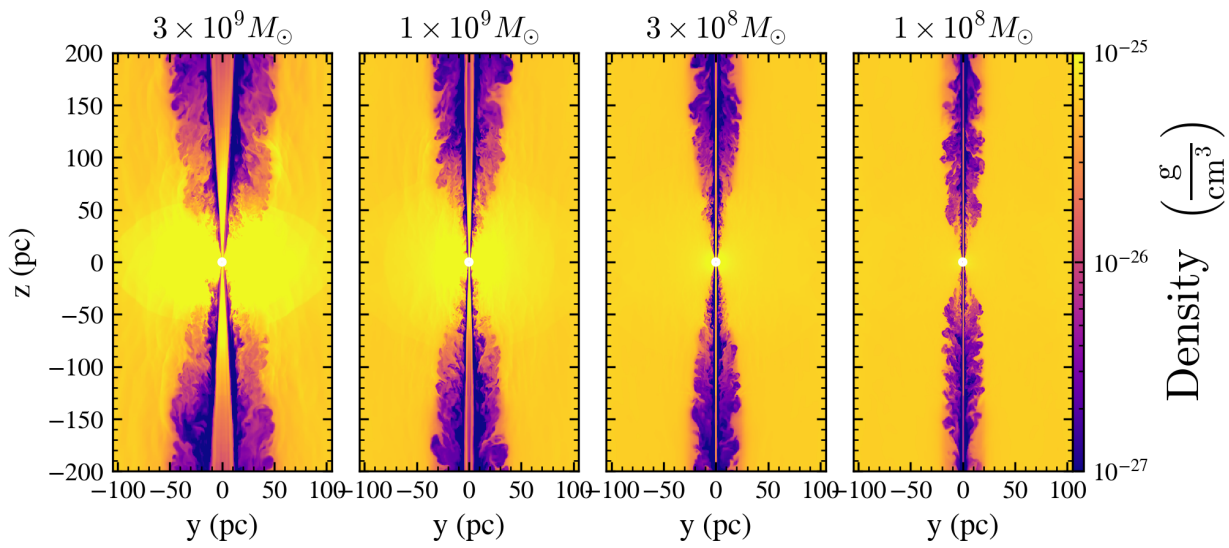

Figure 18. Gas density in the yz plane at $t = 0.2$ Myr for the JD runs, with fixed control surface size ($R_{cs} = 2$ pc) and resolution ($\Delta x = 0.25$ pc) and fixed radius and height of the jet injection cylinder ($r_{jet} = h_{jet} = 0.5$ pc). The mass of the central black hole varies.

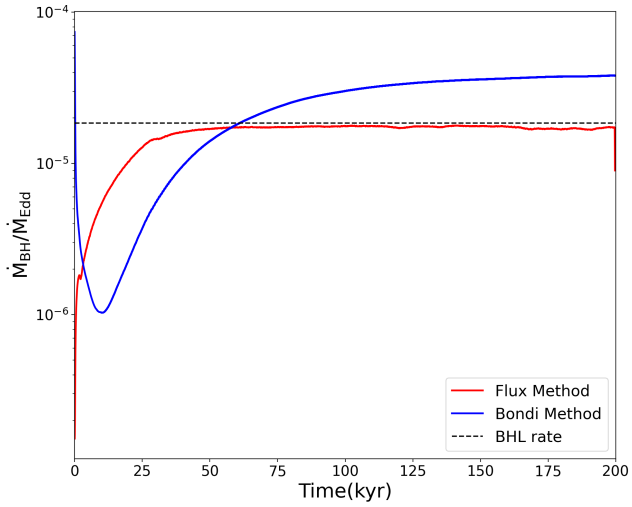

Figure 19. Evolution of accretion rate for different accretion scenarios (with measuring accretion rate using the Bondi formula or using our flux approach, run J0 vs. JC1 in Table 1), with the same initial condition and jet model. We use a fixed control surface size ($R_{cs} = 2$ pc), jet injection region size ($h_{jet} = 2\ \Delta x, r_{jet} = 2\ \Delta x$), and resolution ($\Delta x = 0.25$ pc).

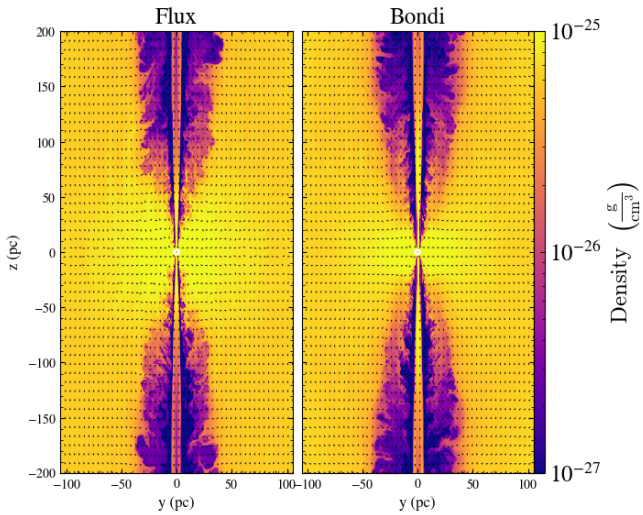

Figure 20. Density slices at same time ($t = 0.2$ Myr) for runs with different accretion mechanisms (the runs shown are the same as Figure 19).

2013; Li et al. 2017). This is also over-simplified, and involves arbitrary parameters that do not have a clear physical meaning and need to be tuned to observations. In our sink particle method, for the accretion model such parameters are taken out, at the cost of requiring higher spatial resolution in the vicinity of the AGN.

In comparison with bubble-type feedback models, our use of a cylindrical window function to impose feedback makes it easier to vary the shape and size of the jet; thus it can be accommodated to different sources and scenarios where the jet dynamics vary, and it is easier to incorporate effects such as precession.

The main deficiency of the jet model presented here is that it still involves an "efficiency parameter," which

is set to be constant and consistent with previous AGN feedback work. We also do not distinguish hot and cold accretion fluxes on the control surface or take the angular momentum of the flow into account in the model.

5. CONCLUSIONS

In this work, we have developed a new subgrid model for accretion and jet feedback in hydrodynamic simulations. The model resembles a sink particle algorithm commonly used in star formation studies, measuring accretion onto SMBHs via a flux across a surface instead of using a simple analytical formula, and incorporating inner boundary conditions which remove the gas accreted. With this formulation we further investigated different ways to inject jets and characterize mechanical feedback and compared the jet dynamics and feedback properties with other smaller scale work to test the convergence of our model. We have found the following:

1. We tested the accretion part of the model with two idealized scenarios: spherically symmetric Bondi accretion and Bondi-Hoyle-Lyttleton accretion through a wind blowing past the BH particle. With a Bondi solution gas profile setup, we ran tests with varying resolutions, as well as varying the accretion measurement radius (control surface radius) and reset radius. We have managed to reproduce a steady-state flow as predicted with our sink particle model using a kernel reset that does not assume the Bondi solution, while maintaining an accretion rate within 0.1% error compared to the Bondi accretion rate.

2. The Bondi-Hoyle-Lyttleton test was conducted with varying Mach numbers for the ambient wind and different resolutions of $r_{\rm BHL}$. As long as $r_{\rm BHL}$ is resolved by two or more zones, we observe better than 10% accuracy in the measured accretion rate. This means that for lower black hole masses or higher Mach numbers, the resolution requirements are more strict. To reliably reproduce the expected accretion rate within a reasonable error threshold, the control surface also needs to be at least two zones across.

3. We further imposed a jet feedback model and studied the evolution of accretion and feedback as well as environmental effects the jet has on an idealized ICM environment. We ran simulations with different resolutions and jet sizes. Overall, properties including accretion rates, feedback energy, shocked volume, and amount of entrained mass show good convergence behavior with resolution. However, to reproduce some of the shock structures and fluid instabilities, a short, skinny jet injection region is favored. We adopt the $r_{\rm jet} = h_{\rm jet} = 1/4R_{\rm cs}$ configuration for future work.

4. Our work has shown consistency with previous jet simulations at smaller scales, while as a standalone model it is able to reproduce the jet dynamics better than many current AGN feedback simulations have done. We are able to see some dynamic jet features that are observed in MHD jet simulations at parsec and kiloparsec scales, as well as instability at the jet-ICM interface.

This model still has several limitations. In the vicinity of the control surface, the numerical jet launching mechanism may interact with the incoming flow in ways that are not fully realistic. In addition, the feedback efficiency parameter described in jet models needs to take into account the accretion disk modeling we discuss in Paper II. Relativistic effects are also not incorporated in the current jet model, but will likely impact the pressure of the shock inflated by the jet.

The model will be used to address some of the key questions regarding cool-core clusters and the cool-core problem in later work.

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Software: FLASH (Fryxell et al. 2000; Dubey et al. 2008), yt (Turk et al. 2011), Matplotlib (Hunter 2007), NumPy (Harris et al. 2020)

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